

MA3236 NONLINEAR PROGRAMMING

Semester 1, AY 2026/2027

Tutorial 1

Review: Lectures 1–2

- **Model and notation**

$$\min_{\mathbf{x} \in S} f(\mathbf{x}) \quad \text{or} \quad \max_{\mathbf{x} \in S} f(\mathbf{x}), \quad \max_S f = -\min_S(-f) \quad (\text{same optimizers}).$$

- $\mathbf{x}$: decision variable; f : objective function; S : feasible set.
- $\mathbf{x} \in S$: feasible; $\mathbf{x} \notin S$: infeasible.

- **Well-posedness for minimization**

$$S \neq \emptyset, \quad \inf_{\mathbf{x} \in S} f(\mathbf{x}) > -\infty, \quad \exists \mathbf{x}^* \in S : f(\mathbf{x}^*) = \inf_{\mathbf{x} \in S} f(\mathbf{x}).$$

- For maximization, replace inf and “bounded below” by sup and “bounded above”.
- S unbounded $\nRightarrow$ the objective is unbounded.

- **Graphical method in $\mathbf{R}^2$**

$$L_c := \{\mathbf{x} \in \mathbf{R}^2 : f(\mathbf{x}) = c\} \quad (\text{level curve}).$$

- Sketch S and representative L_c .
- Minimize: move toward smaller c ; maximize: toward larger c .
- Locate contact points, then check every relevant boundary piece.

- **Local versus global minimizer ($\mathbf{x}^* \in S$)**

$$\begin{aligned} \text{local: } & \exists \varepsilon > 0 : f(\mathbf{x}^*) \leq f(\mathbf{x}) \quad \forall \mathbf{x} \in S \cap B(\mathbf{x}^*, \varepsilon), \\ \text{global: } & f(\mathbf{x}^*) \leq f(\mathbf{x}) \quad \forall \mathbf{x} \in S. \end{aligned}$$

- global $\implies$ local; the converse need not hold.
- Strict: replace $\leq$ by $<$ for every other relevant point.

- **Closedness, compactness, and Weierstrass**

$$\begin{aligned} g, h \text{ continuous} & \implies \{\mathbf{x} : g(\mathbf{x}) = 0\}, \{\mathbf{x} : h(\mathbf{x}) \leq 0\} \text{ are closed,} \\ S \text{ compact} & \iff S \text{ closed and bounded} \quad (S \subseteq \mathbf{R}^n), \\ S \neq \emptyset, S \text{ compact, } f \text{ continuous} & \implies f \text{ attains both } \min_S f \text{ and } \max_S f. \end{aligned}$$

- **Coercivity and existence**

$$f \text{ coercive} \iff \|\mathbf{x}\| \rightarrow \infty \implies f(\mathbf{x}) \rightarrow +\infty.$$

$$S \neq \emptyset, S \text{ closed, } f \text{ continuous and coercive} \implies f \text{ attains } \min_S f.$$

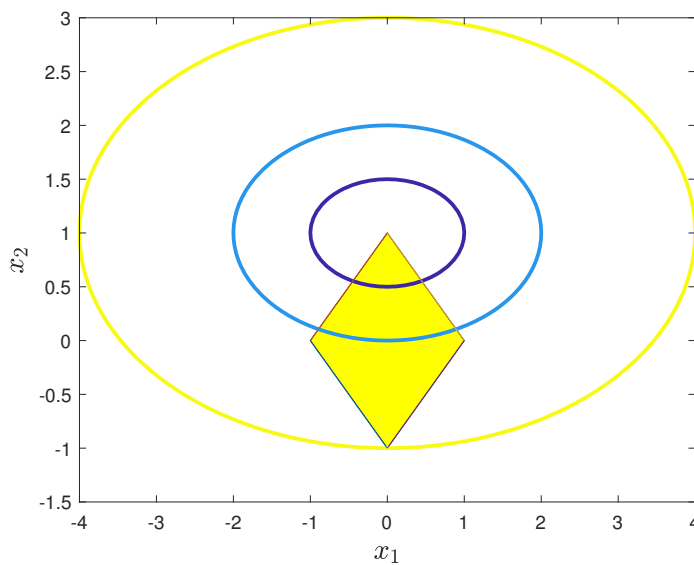
- To disprove coercivity: find $\|\mathbf{x}^k\| \rightarrow \infty$ while $f(\mathbf{x}^k)$ does not tend to $+\infty$.
- Weierstrass and coercivity give sufficient, not necessary, conditions.

1. Consider the following constrained nonlinear programming problem.

$$\begin{aligned} \max \quad & \frac{(x_1)^2}{4} + (x_2 - 1)^2 \\ \text{s.t.} \quad & -1 \leq x_1 + x_2 \leq 1, \\ & -1 \leq x_1 - x_2 \leq 1. \end{aligned}$$

- (a) Sketch the feasible region S .
- (b) Solve the NLP graphically.

Solution.



The optimal solution is $(0, -1)$.

2. Consider the following nonlinear programming problem.

$$\begin{aligned} \text{maximize} \quad & f(\mathbf{x}) := \frac{x_1^2}{4} + x_2^2 \\ \text{subject to} \quad & x_1 + x_2 \leq 2 \\ & x_1 \geq 0, x_2 \geq 0 \end{aligned}$$

- (a) Sketch the feasible set S .
- (b) Sketch the level curves correspond to $f(\mathbf{x}) = 1$ and $f(\mathbf{x}) = 9/4$. Hence determine the ‘direction’ where $f(\mathbf{x})$ increases.
- (c) Solve the NLP.

Solution. (a) $S = \{\mathbf{x} : x_1 + x_2 \leq 2, x_1, x_2 \geq 0\}$ is the region inside the triangle with vertices at $[0; 0]$, $[2; 0]$, and $[0; 2]$.

(b) See Figure 1. $f(\mathbf{x})$ increases outwards from the origin.

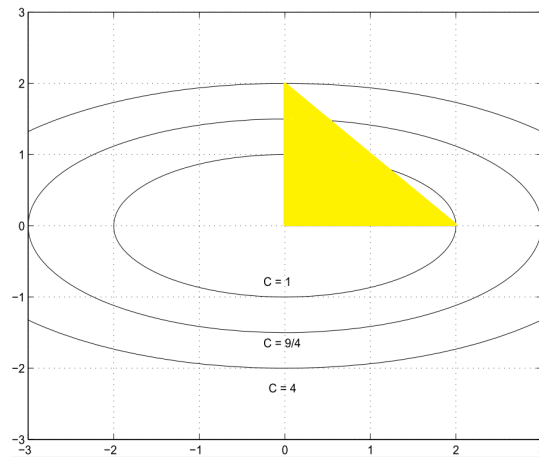


Figure 1: Contours $\{\mathbf{x} \in \mathbf{R}^2 : f(\mathbf{x}) = C\}$. Contour values are $C = 1, 9/4, 4$.

(c) From Figure 1, the contour with the largest function value that last touches the feasible region S is the contour with value 4. It touches S at the point $\mathbf{x} = [0; 2]$. Thus $\mathbf{x} = [0; 2]$ is the global maximizer.

Note. Since S is closed and bounded and f is continuous, the Weierstrass Theorem guarantees that both a global maximizer and a global minimizer exist.

3. Consider the following constrained nonlinear programming problem.

$$\begin{aligned} & \text{maximize} && x_1 + x_2 \\ & \text{subject to} && -x_1^2 + x_2 \leq 0, \\ & && -2 \leq x_1 \leq 2. \end{aligned}$$

- (a) Sketch the feasible region S .
- (b) Identify all local maximizers graphically.
- (c) Is there a global maximizer? If yes, determine all global maximizers.
- (d) Is it true that a local maximizer is a global maximizer? Justify your answer.

Solution. (a) See Figure 2.

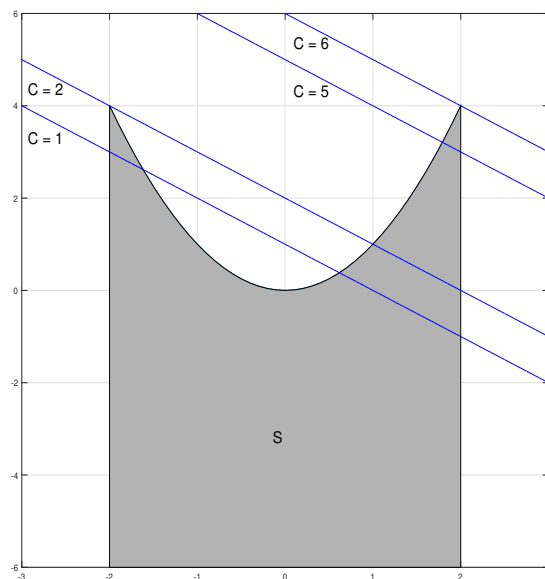


Figure 2: Contours $\{\mathbf{x} \in \mathbf{R}^2 : x_1 + x_2 = C\}$.

(b) From the contours in Figure 2, we see that in a local neighborhood of the point $[-2; 4]$, the function value $f(\mathbf{x}) = 2$ is locally the maximum. Similarly for the point $[2; 4]$.

(c) and (d). Since the contour with the largest value touches the feasible region at the point $\mathbf{x} = [2; 4]$, it is the global maximizer. It is not necessary for a local maximizer to be also a global maximizer, see for example the point $\mathbf{x} = [-2; 4]$.

4. For each of the following NLPs, determine whether there is an optimal solution. Justify your answers. You need not find the optimal solution explicitly.

(a)

$$\begin{aligned} \min \quad & f(\mathbf{x}) := \frac{x_1^4 + x_1^3 x_2 + x_2^4 + 3}{x_1^4 + x_1 x_2^3} \\ \text{s.t.} \quad & (x_1 - 2)^2 + x_2^2 \leq 1 \\ & x_2 \geq 0 \end{aligned}$$

(b)

$$\begin{aligned} \min \quad & g(\mathbf{x}) := x_1^2 x_2 \\ \text{s.t.} \quad & x_1^2 - x_2^2 \leq 4 \\ & x_2 \geq 0 \end{aligned}$$

(c)

$$\begin{aligned} \min \quad & h(\mathbf{x}) := x_1 - x_2 \\ \text{s.t.} \quad & x_1^2 - x_2^2 = 4 \\ & x_2 \geq 0 \end{aligned}.$$

(d)

$$\begin{aligned} \min \quad & q(\mathbf{x}) := x_1^4 + x_2^4 - x_1^2 x_2^2 - 2x_1^3 + x_2 \\ \text{s.t.} \quad & x_1 + x_2 \geq 1 \end{aligned}.$$

Solution. (a) The feasible set S is closed and bounded (it is the upper-half part of the disk centred at $(2,0)$), and f is continuous on S (note that the denominator of f is non-zero on S). By Weierstrass Theorem, the NLP has a global minimizer.

Note: it also has a global maximizer by the Weierstrass Theorem.

(b) Let S be the feasible region. Since $x_2 \geq 0$ and $x_1^2 \geq 0$, we have $g(\mathbf{x}) = x_1^2 x_2 \geq 0$ for all $\mathbf{x} \in S$. Note that $\mathbf{0} \in S$, and $g(\mathbf{0}) = 0$. Thus $\mathbf{0}$ is a global minimizer of the NLP.

Note: although S is unbounded, g is bounded below and the lower bound is attainable by the minimizer $\mathbf{0}$.

(c) The feasible region is the set

$$S = \{\mathbf{x} : x_2 = \sqrt{x_1^2 - 4}, x_1 \geq 2\} \cup \{\mathbf{x} : x_2 = -\sqrt{x_1^2 - 4}, x_1 \leq -2\}.$$

Thus $h(\mathbf{x}) = x_1 - \sqrt{x_1^2 - 4}$ for $\mathbf{x} \in S$. Now $h(\mathbf{x}) \rightarrow -\infty$ as $x_1 \rightarrow -\infty$. Thus the NLP does not have a global minimizer.

(d) Let

$$S := \{\mathbf{x} \in \mathbf{R}^2 : x_1 + x_2 \geq 1\}.$$

The set S is nonempty and closed. Moreover,

$$x_1^2 x_2^2 \leq \frac{1}{2}(x_1^4 + x_2^4).$$

If $r := \max\{|x_1|, |x_2|\}$, then

$$\begin{aligned} q(\mathbf{x}) &\geq \frac{1}{2}(x_1^4 + x_2^4) - 2|x_1|^3 - |x_2| \\ &\geq \frac{1}{2}r^4 - 2r^3 - r \longrightarrow +\infty \quad (r \rightarrow \infty). \end{aligned}$$

Since $\|\mathbf{x}\| \rightarrow \infty$ if and only if $r \rightarrow \infty$, the objective q is coercive. It is also continuous. Hence the existence theorem for continuous coercive functions on nonempty closed sets guarantees that the NLP has a global minimizer.